



TECNOLÓGICO
NACIONAL DE MÉXICO



Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Computo en la nube

4.1 Práctica de seguridad informática escalada de privilegios

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05/03/2026

Creación de usuario y contraseña

```
jesusc@SRVREP:~$ sudo adduser usuario_test
[sudo] password for jesus:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for usuario_test
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
jesusc@SRVREP:~$ sudo passwd usuario_test
New password:
Retype new password:
passwd: password updated successfully
jesusc@SRVREP:~$
```

Agregamos vulnerabilidad editando el archivo de usuarios

```
jesusc@SRVREP: ~  
GNU nano 8.4 /etc/sudoers.tmp *  
# equivalent users (group sudo)  
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"  
  
# This allows running arbitrary commands, but so does ALL, and it means  
# different sudoers have their choice of editor respected.  
#Defaults:%sudo env_keep += "EDITOR"  
  
# Completely harmless preservation of a user preference.  
#Defaults:%sudo env_keep += "GREP_COLOR"  
  
# While you shouldn't normally run git as root, you need to with etckeeper  
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"  
  
# Per-user preferences; root won't have sensible values for them.  
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"  
  
# "sudo scp" or "sudo rsync" should be able to use your SSH agent.  
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"  
  
# Ditto for GPG agent  
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"  
  
# Host alias specification  
  
# User alias specification  
  
# Cmnd alias specification  
  
# User privilege specification  
root    ALL=(ALL:ALL) ALL  
  
# Allow members of group sudo to execute any command  
%sudo    ALL=(ALL:ALL) ALL  
  
# See sudoers(5) for more information on "@include" directives:  
  
@includedir /etc/sudoers.d  
  
usuario_test ALL=(ALL) NOPASSWD: /usr/bin/nano|  
  
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location  
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Entramos con el usuario prueba y modificamos el archivo /etc/passwd que nos permite eliminar la contraseña de root:

```
visudo: /etc/sudoers.tmp unchanged
jesusc@SRVREP:~$ su - usuario_test
Password:
usuario_test@SRVREP:~$ sudo nano /etc/passwd
usuario_test@SRVREP:~$ su root
root@SRVREP: /home/usuario_test# sudo visudo
```



```
usuario_test@SRVREP: ~
GNU nano 8.4 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:991:991:System Message Bus:/nonexistent:/usr/sbin/nologin
jesusc:x:1000:1000:jesusc,,,:/home/jesusc:/bin/bash
sshd:x:990:65534:sshd user:/run/sshd:/usr/sbin/nologin
mysql:x:101:105:MariaDB Server:/nonexistent:/bin/false
usuario_test:x:1001:1001:,,,:/home/usuario_test:/bin/bash

[ Read 25 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

Eliminamos la contraseña de root:

```
usuario_test@SRVREP:~$ sudo nano /etc/passwd
usuario_test@SRVREP:~$ su root
root@SRVREP:/home/usuario_test# |
```

Eliminamos la línea en el archivo de usuarios que creo la vulnerabilidad:

```
usuario_test@SRVREP: ~
GNU nano 8.4 /etc/sudoers.tmp *
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

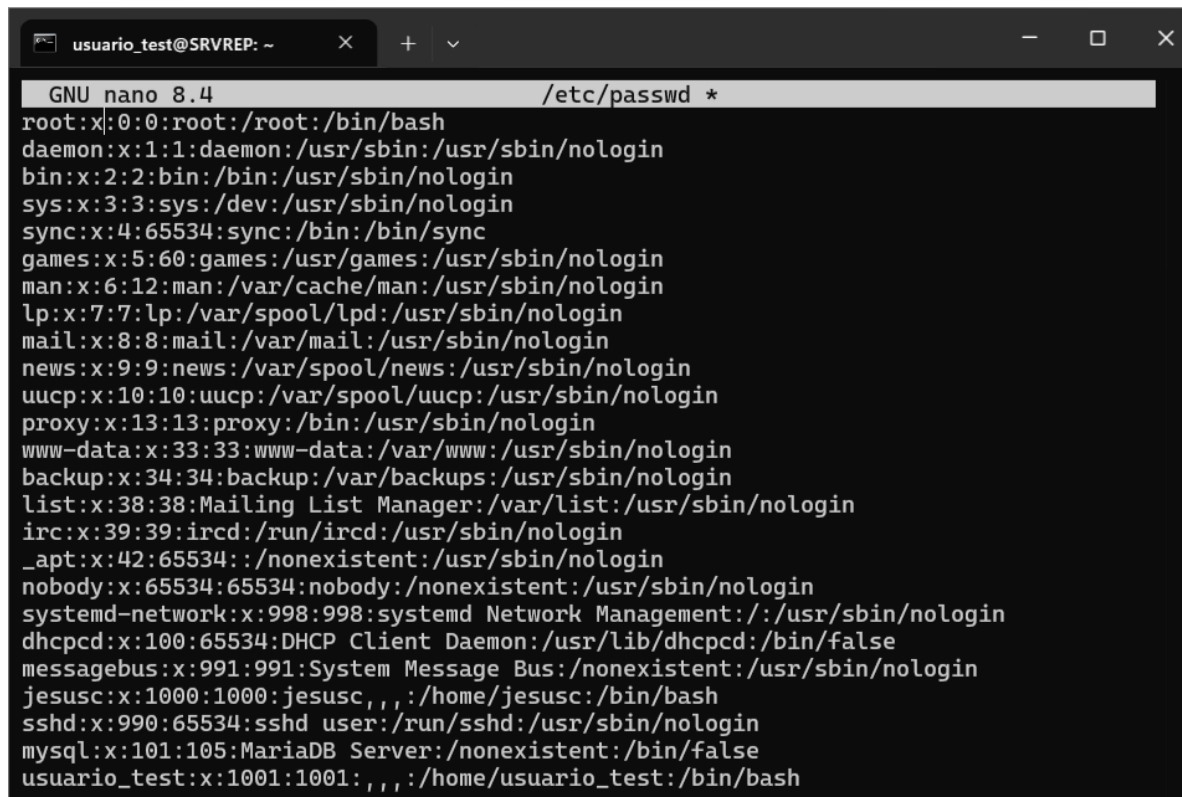
# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@include /etc/sudoers.d

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Restauramos la línea que elimina la contraseña de root:



The screenshot shows a terminal window with the title 'usuario_test@SRVREP: ~'. Inside, the GNU nano 8.4 editor is open, displaying the contents of the /etc/passwd file. The file lists system users and regular users, each with their username, UID, GID, name, home directory, and shell. The root user is listed with UID 0 and shell /bin/bash. The usuario_test user is at the bottom with UID 1001 and shell /bin/bash.

```
GNU nano 8.4 /etc/passwd *
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:991:991:System Message Bus:/nonexistent:/usr/sbin/nologin
jesusc:x:1000:1000:jesusc,,,:/home/jesusc:/bin/bash
sshd:x:990:65534:sshd user:/run/sshd:/usr/sbin/nologin
mysql:x:101:105:MariaDB Server:/nonexistent:/bin/false
usuario_test:x:1001:1001:,,,:/home/usuario_test:/bin/bash
```

Con eso concluye la práctica, ésta me ha permitido identificar el cómo podemos hacer para mitigar vulnerabilidades dentro de Linux.

```
jesusc@SRVREP:~$ sudo adduser usuario_test
[sudo] password for jesus:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for usuario_test
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
jesusc@SRVREP:~$ sudo passwd usuario_test
New password:
Retype new password:
passwd: password updated successfully
jesusc@SRVREP:~$ sudo visudo
jesusc@SRVREP:~$ sudo visudo
visudo: /etc/sudoers.tmp unchanged
jesusc@SRVREP:~$ su - usuario_test
Password:
usuario_test@SRVREP:~$ sudo nano /etc/passwd
usuario_test@SRVREP:~$ su root
root@SRVREP:/home/usuario_test# sudo visudo
root@SRVREP:/home/usuario_test# sudo nano /etc/passwd
root@SRVREP:/home/usuario_test# su - jesus
jesusc@SRVREP:~$ |
```